

Guide: Which Late Bridges Cannot Work

Collatz proof project — September 5, 2026

The search being studied

Our transfer search follows a known-convergent source and occasionally uses a smaller transfer instance to introduce an auxiliary source. At the level of base values, the bridge replaces x by $9x + 2$. The aim is to make a source meet the target uniformly across a whole arithmetic progression.

The stronger result

Lean now proves a size bound for every finite path combining ordinary forward Collatz steps and these bridges, in any order. The proof does not limit the number of bridges or require them to occur at a particular value. It even allows more bridges than the actual induction rules permit, so its impossibility consequence applies to the restricted search too.

Why the bound is simple

An even step halves the value. An odd shortcut step increases it by at most a factor of two. A bridge increases it by at most a factor of sixteen. Multiplying those bounds gives a constraint on the endpoint after any combination of operations. Lean checks the multiplication argument for arbitrary paths.

What happens at the trivial cycle

Once a base value is 1 or 2, we can combine the clock, its cycle phase, and the symbolic slope exponent into an integer quantity called its charge. Ordinary steps around the cycle preserve this charge. The new bound proves that a source cannot lower its charge by leaving the cycle through bridges and eventually returning, even if it inserts more bridges during the excursion.

If the source charge is already above the target charge, it cannot match the target later using these forward operations. Increasing the future bridge budget will not fix that mismatch.

How this improves the search

The program now discards such states immediately. A complete comparison for parameters 0 through 1000 found the same certificate outcomes and first successful depths with this pruning switched on or off. All eight affine-search tests passed. The pruning rule rests on a general Lean inequality, not on the finite comparison alone.

What remains possible

The result does not exclude auxiliary paths introduced earlier, inverse orbit steps, or different proof rules. It also does not say that the excluded parameters fail to converge. They are exclusions from this particular search strategy after it reaches a particular state.

The full Collatz goal still needs a proof covering every parameter. These findings tell us where adding more of the same forward bridges cannot help; they do not supply the missing global argument.

Where the proof lives

Build `Collatz.BridgeGrowth` from `lean/`. Its three public results are included in the theorem axiom audit. The technical paper gives the exact inequality and its application to symbolic endpoint matching.